

CE 310 — Engineering Systems Analysis

Pre-Lab Quiz: Week 2 — Shape Matters: Reading Distributions

Due: Before Week 2 Lab (Class 3) — submit via D2L Total: 10 points (1 point each)
Estimated time: ~10 minutes

Instructions: Select the single best answer for each question. This quiz is auto-graded and must be completed before attending the Week 2 lab session. You may use your notes and readings, but work independently.

Questions

Q1. In February 2021, over 40 Texas power plants failed during an extreme cold snap, causing blackouts that killed hundreds of people. Power companies had planned capacity based on historical average winter temperatures. Which statistical concept did their planning fail to account for?

- A) The median winter temperature
 - B) The standard deviation of summer temperatures
 - C) The left tail of the temperature distribution — rare but extreme cold events
 - D) The correlation between temperature and wind chill
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Q2. A histogram has its tallest bar (mode) on the right side, with a long tail stretching toward lower values on the left. This distribution is best described as:

- A) Right-skewed (positively skewed)
 - B) Left-skewed (negatively skewed)
 - C) Symmetric with zero skewness
 - D) Bimodal
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Q3. An HVAC engineer sizes the dehumidification system for a Tucson office building using the annual average relative humidity (~33%). During the monsoon season, average monthly humidity can exceed 40%. What is the most likely design consequence?

- A) The system will be oversized and waste energy year-round
- B) The system will be correctly sized — annual average is always the appropriate design basis
- C) The system will be undersized, unable to maintain comfort during peak humidity months
- D) The system will automatically switch to a recirculation mode during the monsoon

Q4. In the Tucson 2023 weather dataset you used in Week 1, which month has the highest average relative humidity?

- A) July (start of the North American Monsoon)
- B) August (peak monsoon month)
- C) January (winter Pacific moisture)
- D) December (end of year storms)

Q5. A dataset has $Q1 = 58^{\circ}\text{F}$, $Q3 = 86^{\circ}\text{F}$, and $IQR = 28^{\circ}\text{F}$. According to the Tukey rule used in box-and-whisker plots, the whiskers extend to:

- A) The nearest data value within $1.0 \times IQR$ of the box edge
- B) The nearest data value within $1.5 \times IQR$ of the box edge
- C) The nearest data value within $2.0 \times IQR$ of the box edge
- D) The absolute maximum and minimum values in the dataset

Q6. A dataset of annual wind speeds has a skewness coefficient of $+1.07$. Which statement is correct?

- A) The distribution is left-skewed, and the mean is pulled below the median by rare very-low-wind-speed events
- B) The distribution is right-skewed — most hours have low wind speeds, with a long tail of occasional high gusts — and the mean is pulled above the median by that tail
- C) The distribution is symmetric, so mean = median
- D) The distribution is right-skewed, but skewness alone provides no information about the relative position of the mean and median

Q7. In a histogram of Tucson's 2023 hourly temperatures using 10°F bins, which bin contains the most hours (i.e., which bin is the mode)?

- A) $60\text{--}70^{\circ}\text{F}$
- B) $70\text{--}80^{\circ}\text{F}$
- C) $80\text{--}90^{\circ}\text{F}$
- D) $90\text{--}100^{\circ}\text{F}$

Q8. In machine learning and data science, the systematic process of visualizing feature distributions, identifying skewness and outliers, and understanding relationships between variables before building a model is called:

- A) Cross-validation
- B) Hyperparameter tuning

- C) Exploratory data analysis (EDA)
 - D) Regularization
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Q9. Which Excel formula correctly returns the 95th percentile of temperature values in the range B2:B8761?

- A) =PERCENTILE(B2:B8761, 95)
 - B) =PERCENTILE(B2:B8761, 0.95)
 - C) =LARGE(B2:B8761, 95)
 - D) =QUARTILE(B2:B8761, 95)
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Q10. A student wants to calculate the skewness coefficient of wind speed values in C2:C8761. Which Excel formula is correct?

- A) =SKEWNESS(C2:C8761)
 - B) =SKEW(C2:C8761)
 - C) =SKEW.P(C2:C8761)
 - D) =KURT(C2:C8761)
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